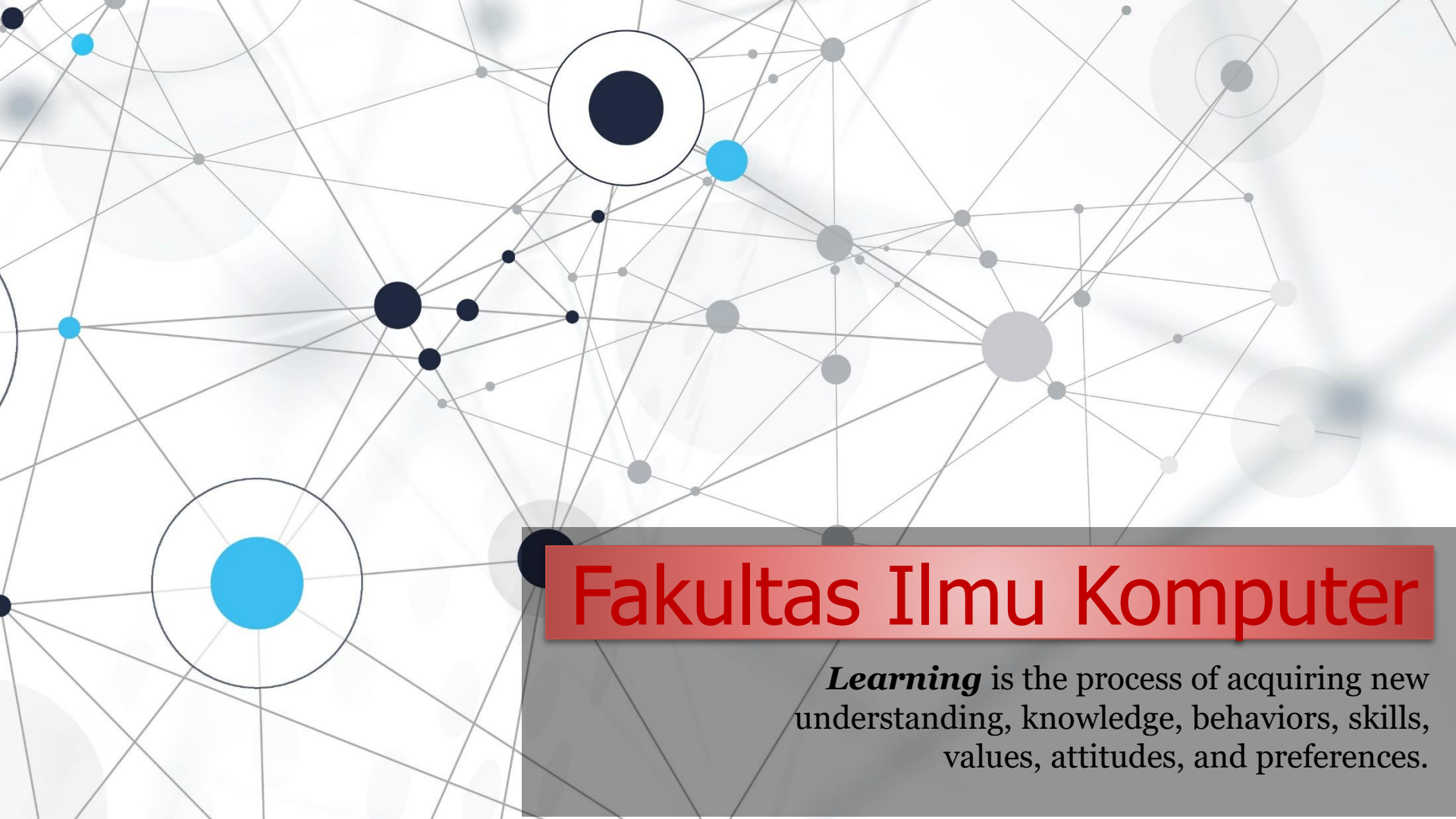


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

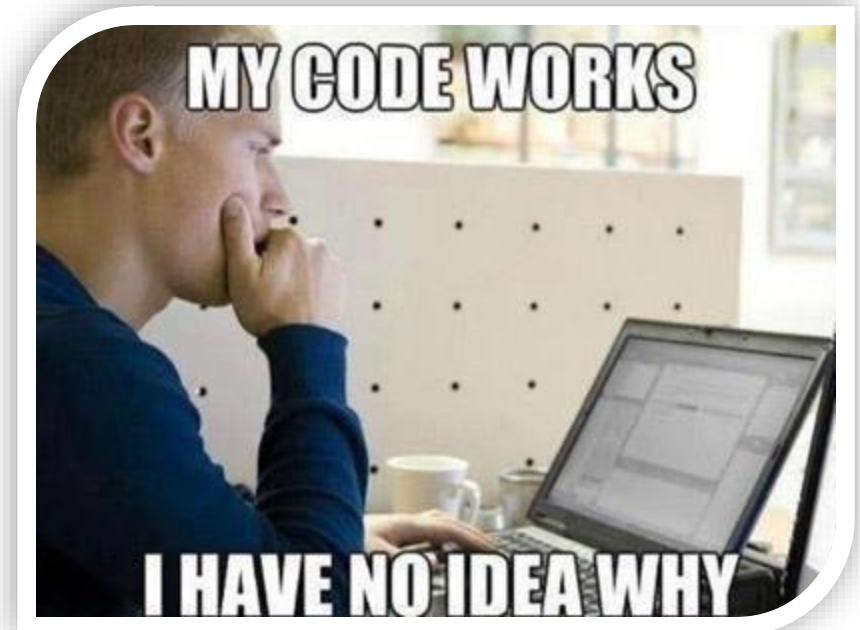
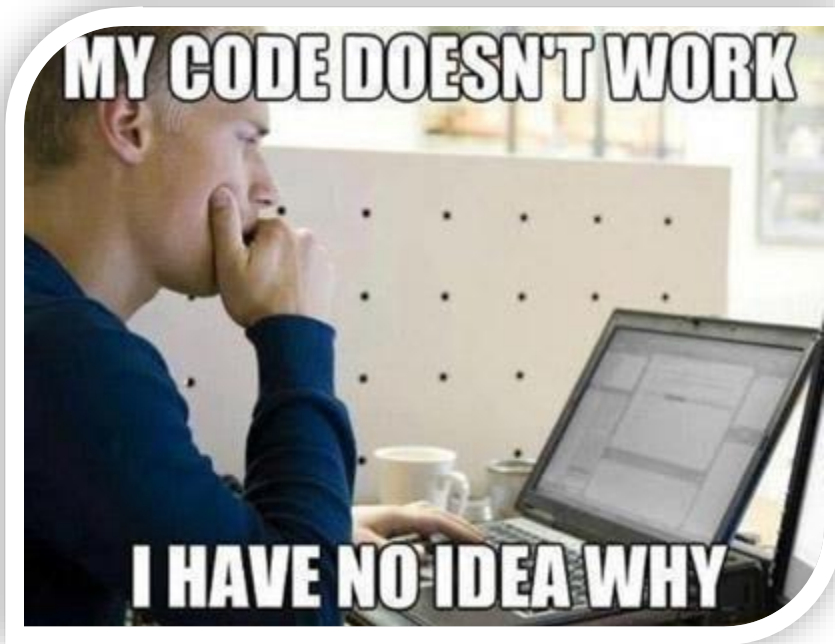
Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Containerization & Configuration Management *in* DevOps

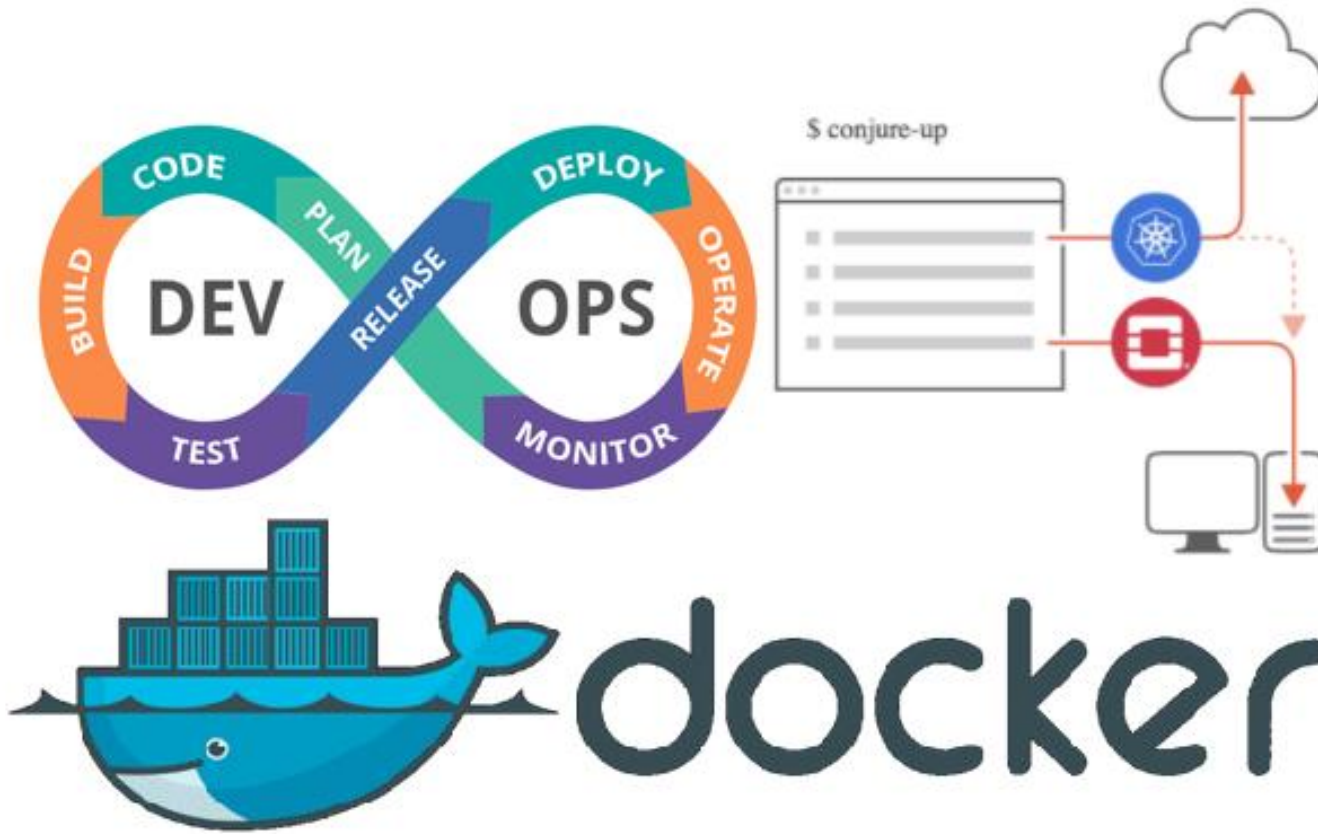


Sudah di EXECUTE di laptop saya dan bisa RUN, tapi mengapa tidak bisa RUN di computer lain?



Seiring dengan pertumbuhan **TECHNOLOGY**, definisi APLIKASI berubah dari hanya **piece of code** menjadi sekumpulan kode (*set of code*), **binari** (*binaries*), **konfigurasi** (*configurations*), dan **environment** yang berjalan (*running environment*).





Containerization, Teknologi yang Lagi Gencar
Digunakan Tahun 2022-2023



Cloud availability in DevOps is a race of all **cloud computing platforms** to ensure that applications and services hosted on them are consistently accessible and functional.

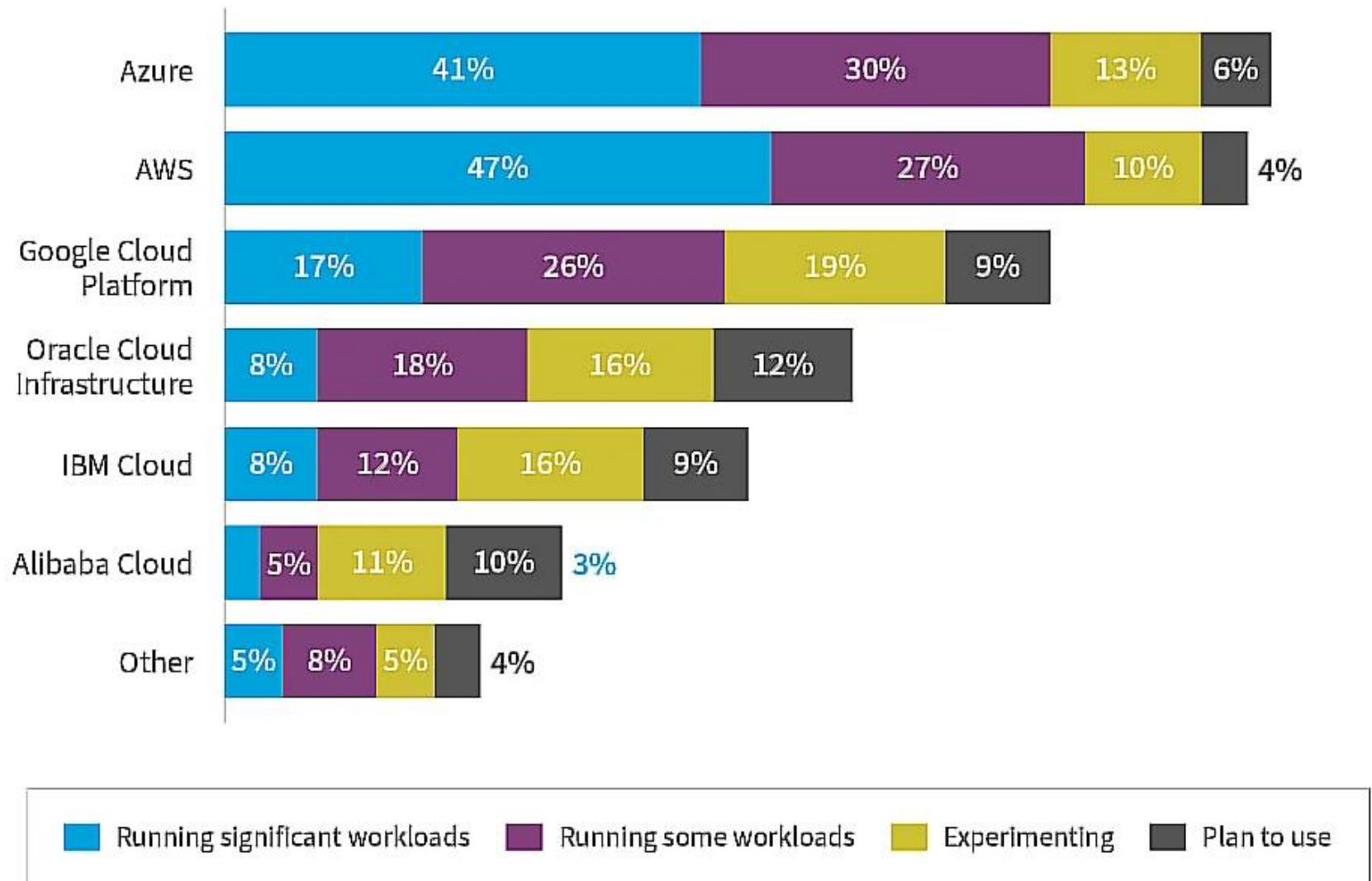
Top DevOps Platforms in 2024:

- 1) Octopus.
- 2) Docker Hub.
- 3) Jenkins.
- 4) Copado.
- 5) Open DevOps by Jira.
- 6) Oracle Cloud.
- 7) Microsoft Azure DevOps Services.
- 8) etc.



DevOps is a methodology that combines cultural philosophies, practices, and tools to **enhance an organization's ability to deliver applications and services** at high velocity. It involves the collaboration and partnership between software development (Dev) and IT operations (Ops) to collaborate through the software lifecycle, from development and testing to deployment to monitoring.

What public cloud providers does your organization use?



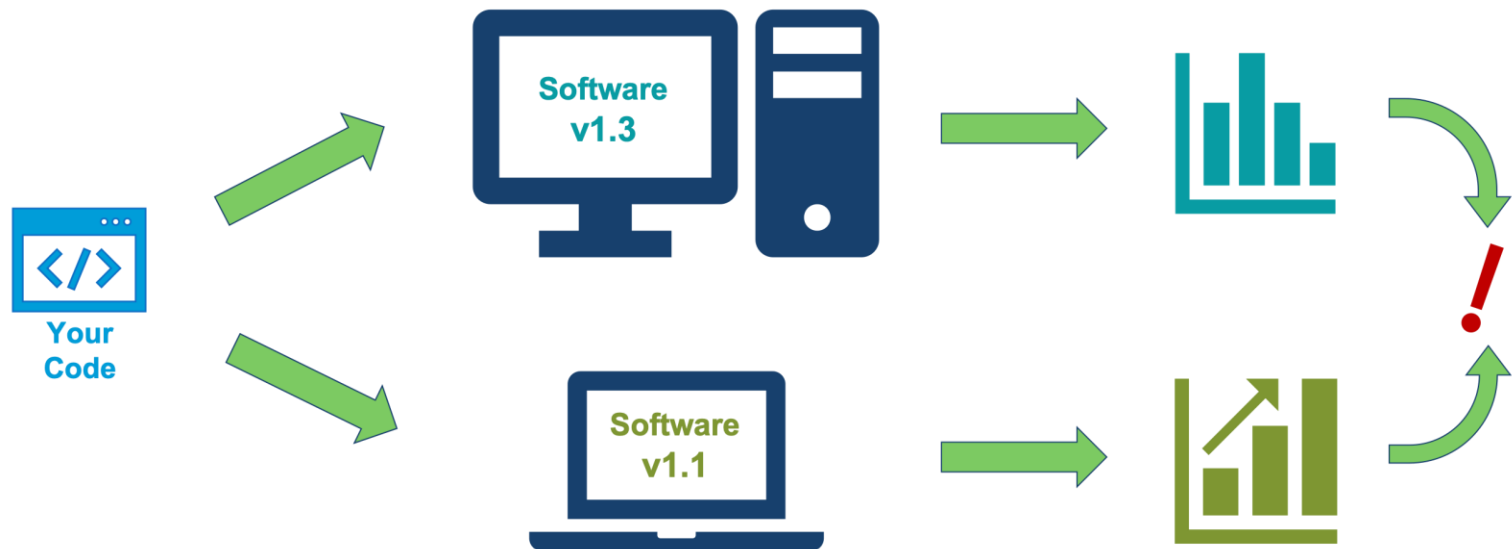
A decorative border of green leaves and branches at the top of the slide.

1 – Containerization *in* DevOps

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Cloud Computing & DevOps

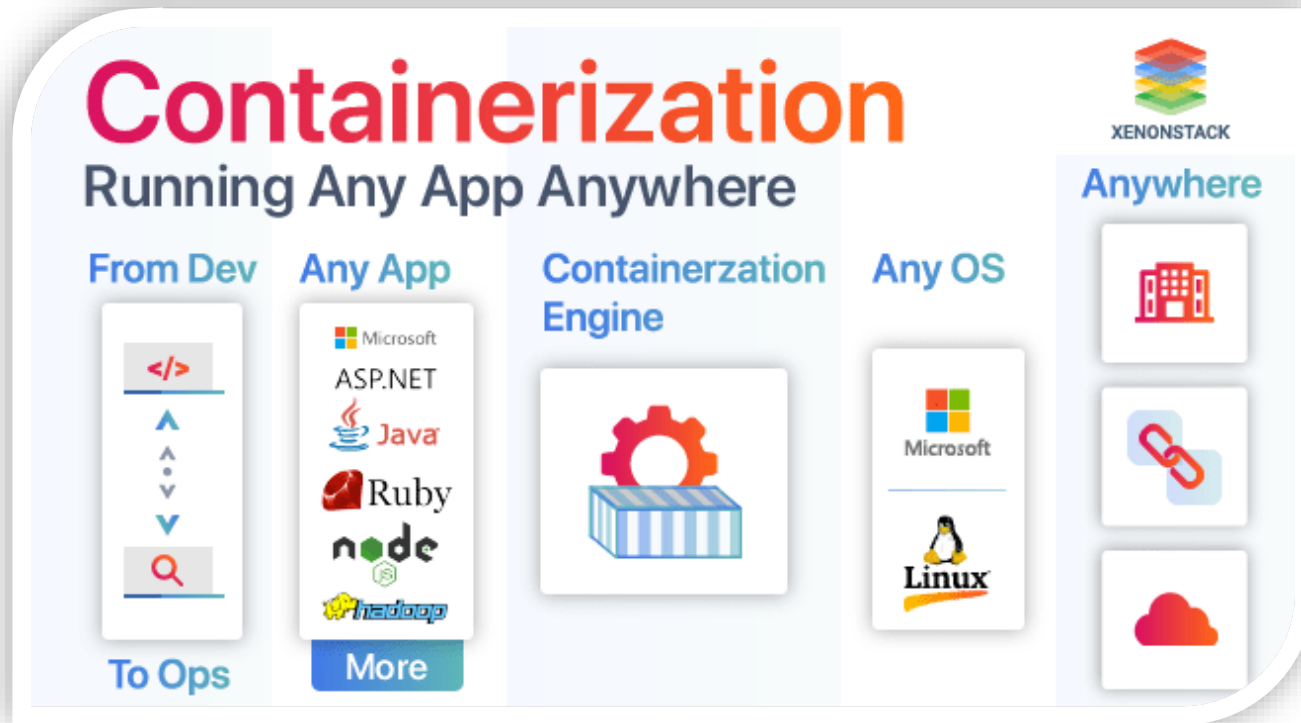
When DevOps teams work in the cloud, they enjoy **easier access** to scalable hardware resources that can help build, test and deploy new updates and offerings more quickly.



What is Containerization in DevOps?

- ❖ **Containerization** is when developers bundle a program's code and all of the other components it needs to run into a **single package**. The advantage is that containerized applications need fewer resources to run and are easy to deploy in new environments.
- ❖ **Containerization** allows businesses to virtualize the operating system and install applications in isolated regions called containers. Containerization is a modern way of creating applications.

Containerization adalah bentuk virtualisasi di mana aplikasi berjalan di ruang user yang terisolasi, yang disebut sebagai container, saat menggunakan sistem operasi bersama (OS) yang sama. Salah satu manfaat containerization adalah bahwa container pada dasarnya adalah lingkungan komputasi yang dikemas penuh dan portabel.



❖ What is Container ???

- ❖ **Containers** are like standard unit or package that hold everything an application needs such as its **dependencies, resources, libraries, source code**, and many more. This helps to achieve portability and reduction in size.
- ❖ **Portability** means the software or system can be transferred or adapted to different *environments, platforms, or operating systems* without significant modification.
- ❖ A **container** is a standard unit of software that **packages up code and all its dependencies**, so the application runs quickly and reliably from one computing environment to another.

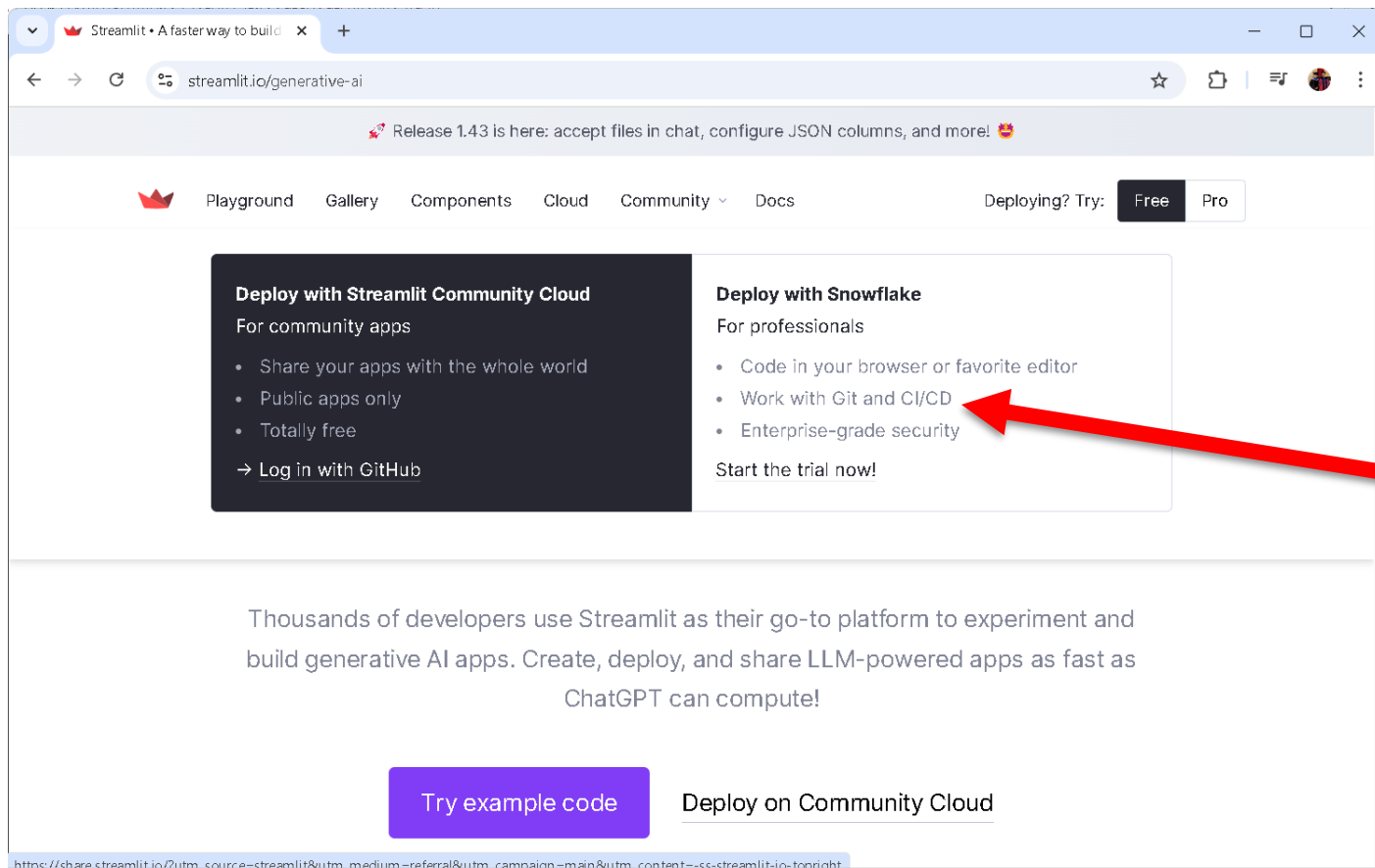
Containerization Concept *in* DevOps

- ❖ **Containerization** entails/includes placing a **software component and its environment, dependencies, and configuration, into an isolated unit called a container**. This makes it possible to deploy an application consistently on any *computing environment*, whether on-premises or cloud-based.
- ❖ **Docker containers** are a natural fit for DevOps, providing significant advantages compared to virtualization or bare-metal deployment. They are easier and faster to deploy, require fewer resources to run, are easier to manage, and are generally more flexible.
- ❖ **These containerization advantages help DevOps teams break applications into microservices**, each of which can be rapidly updated and deployed, increasing development velocity and improving agility.
- ❖ In addition, **containers allow DevOps teams to standardize the way applications are packaged, delivered and deployed** across the development lifecycle.

AI URL :

<https://streamlit.io/generative-ai>

Streamlit adalah framework open-source untuk membuat aplikasi web interaktif dengan **Python**. Kita dapat mengintegrasikannya dengan model AI seperti yang ada di Hugging Face atau PyTorch.

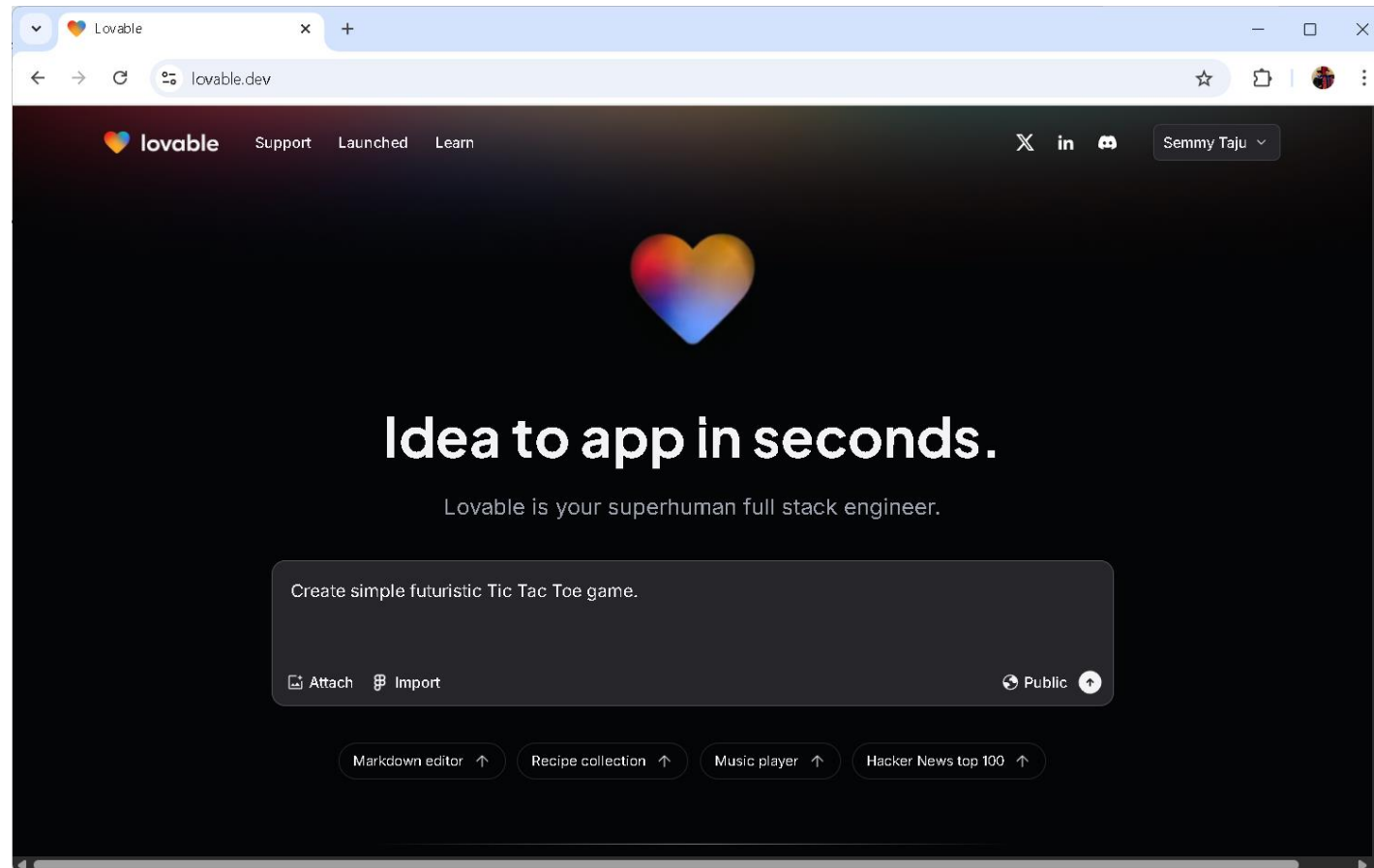


Support
CI/CD

AI URL :

<https://lovable.dev/>

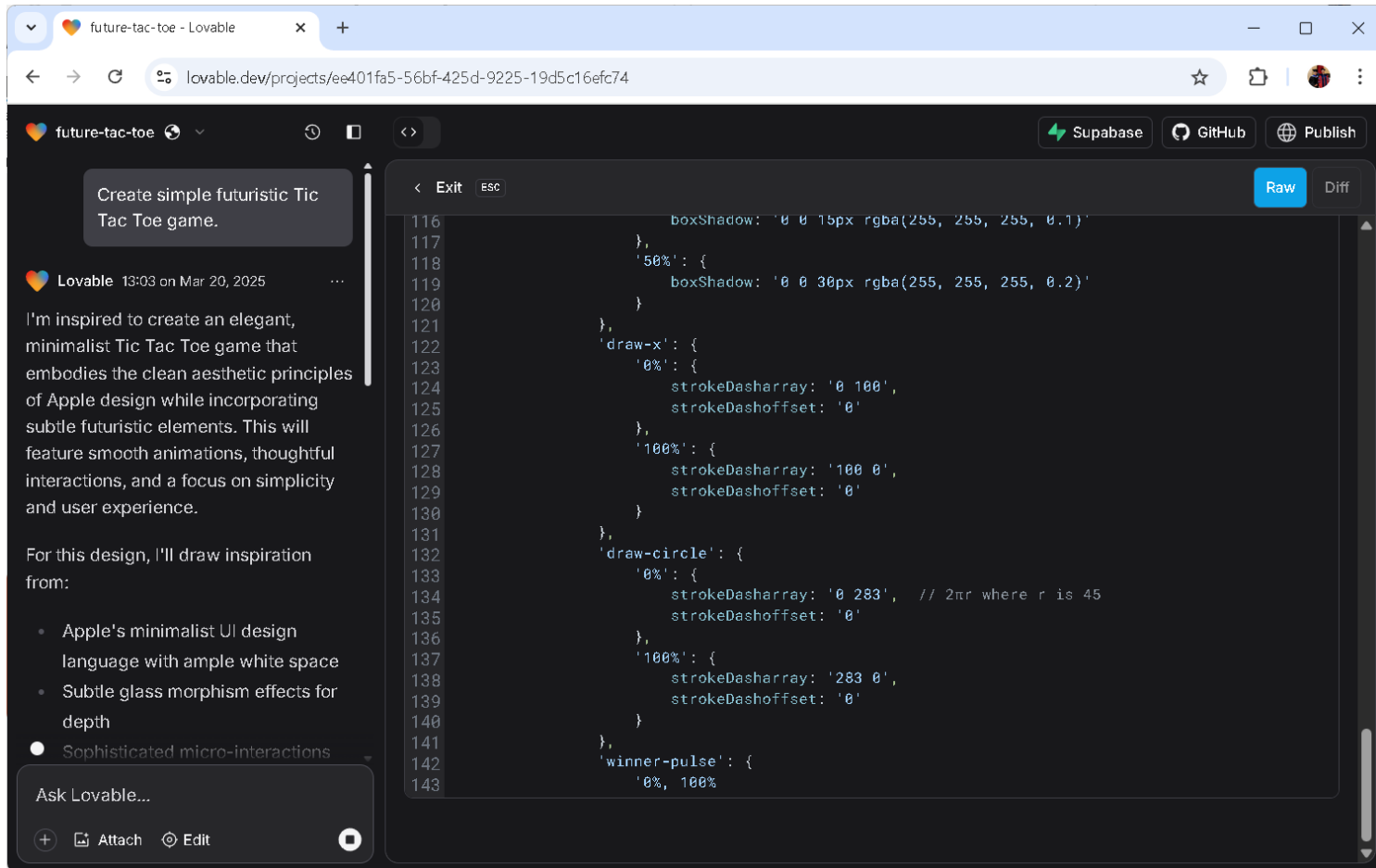
Superhuman full stack engineer



Kalian menuliskan prompt yang sesuai.

AI URL :

<https://lovable.dev/>

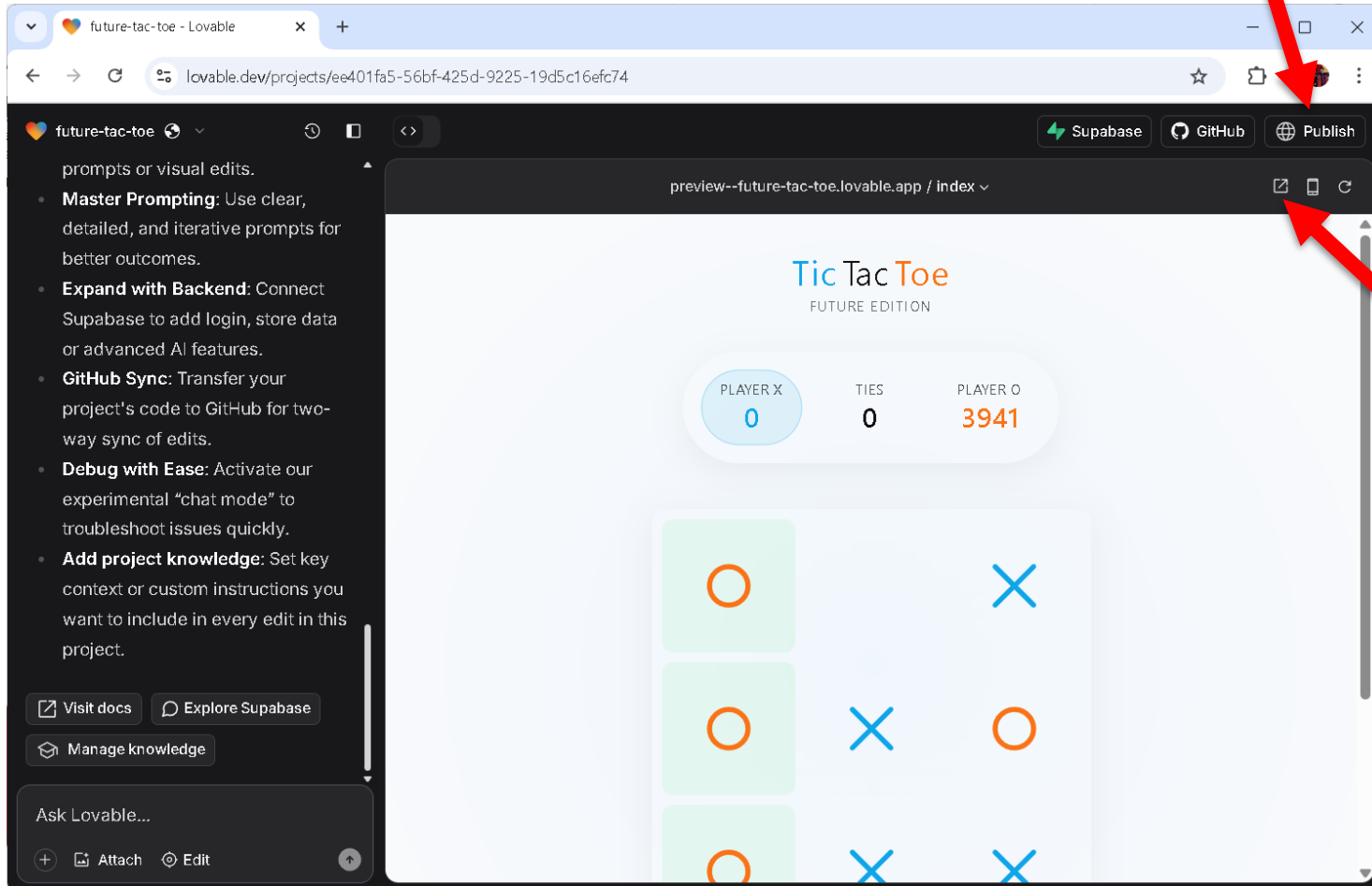


AI akan buatn code program sesuai dengan instruksi.

AI URL :

<https://lovable.dev/>

Click di bagian ini
untuk publish.

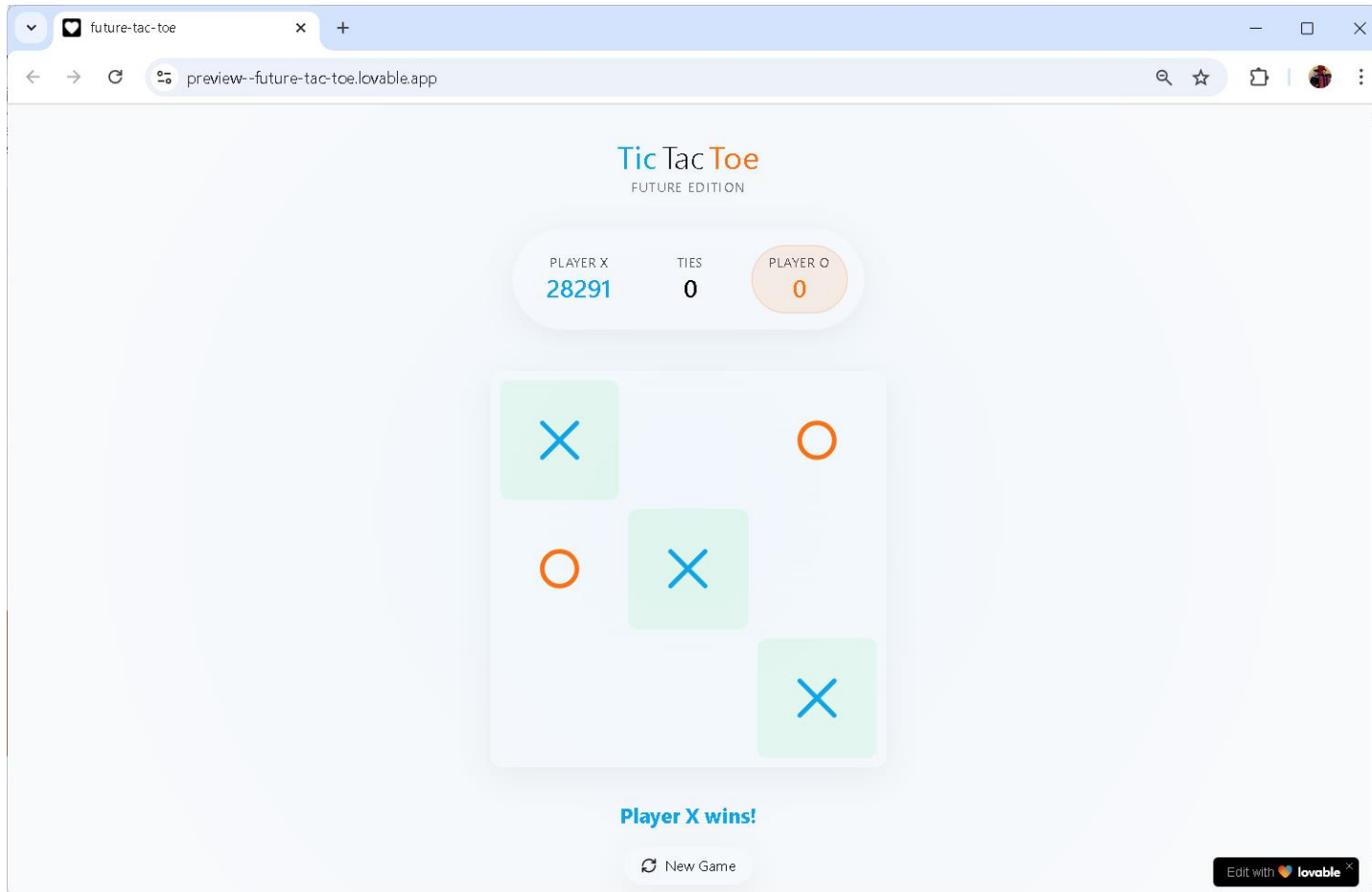


Preview

Code dan simple game telah dibuat, kalian juga bisa koneksi ke database.

AI URL :

<https://lovable.dev/>

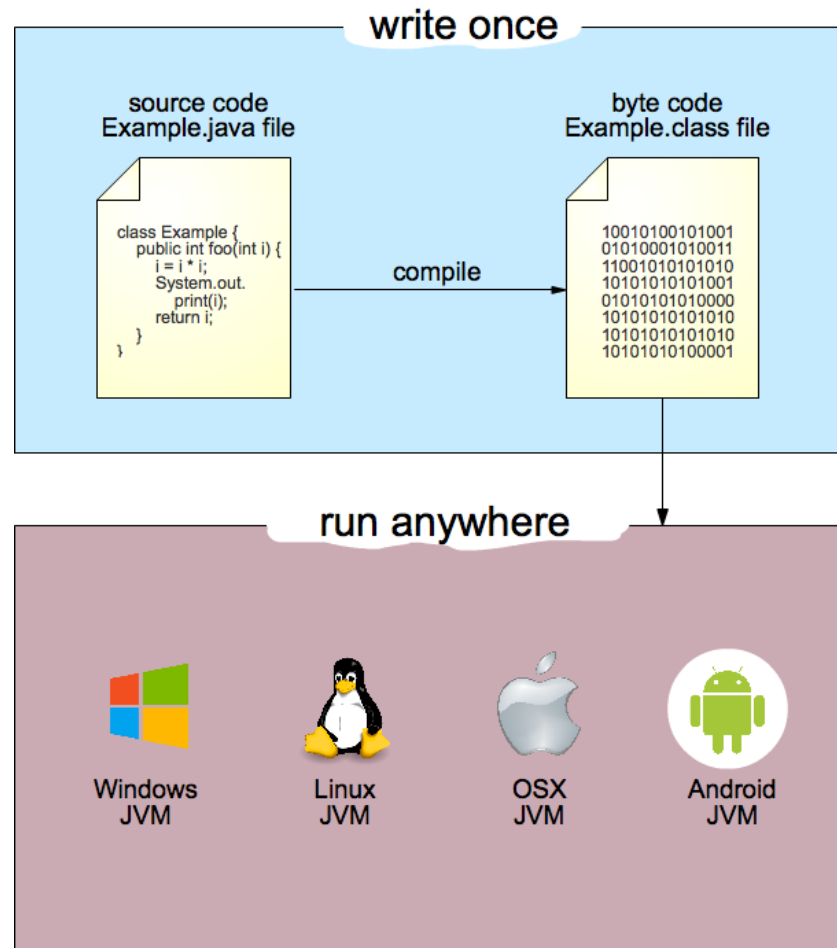


Semuanya sudah bisa mencoba game ini, lakukan testing secara mandiri via URL dibawah:
<https://preview--future-tac-toe.lovable.app/>

Can Containers Benefit DevOps Teams?

- ❖ Containers are a fundamental component of many DevOps processes. They are lightweight, can be deployed consistently in multiple environments, and are easy to transfer from one team to another. In this way, they help foster cross-organizational collaboration.
- ❖ The transition to containers is helping developers and security teams address issues earlier in the development process before they become issues in production environments (a concept known as “shift left security”). Many organizations are fostering collaboration between DevOps and security teams to address security concerns from the onset of the development lifecycle – a pattern known as DevSecOps.
- ❖ **Containers promote shift left because they typically support a single, isolated service, which is easier to automatically test, troubleshoot, and fix throughout the development lifecycle.** A containerized model allows issues to be fixed at the container level without having to re-architect and redeploy entire applications, which can be too complicated and too resource-intensive for agile DevSecOps methodologies.

The integration of software development and IT operations requires rapid change while keeping the cost of change low. IT teams can identify inefficiencies and shift priorities faster.



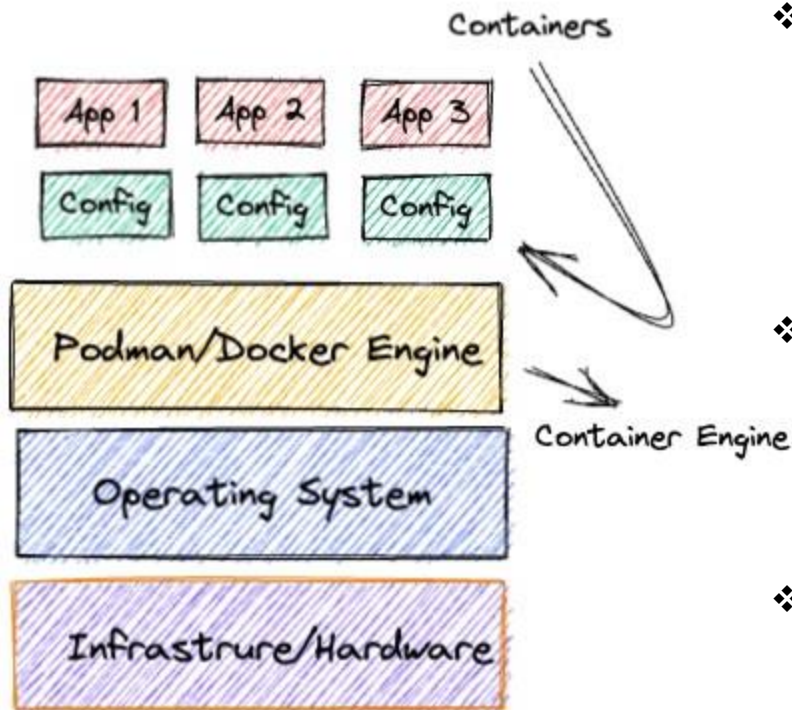
Containers help developers to help their applications run in any environment and are independent of the host hardware. Containers have the specific purpose of being built once and run anywhere.

Containerization vs Virtualization

Property	Containerization	Virtualization
Environment	Containers are packaged with an OS that runs in multiple environments.	Virtualization is built on top of the host, separated by OS appears as a machine.
Startup	Containers take very less to little time on startup.	Virtual Machines take a few minutes to startup.
Resource	These are a minuscule environment that is not at all resource-heavy.	VMs are resource-heavy with no scaling capability.
Implementation	The underlying hardware is virtualized by hypervisors (use of the same hardware).	Containers make the operating system virtual (use of the same OS).
Cost	Easier and inexpensive to implement.	These are expensive and heavy bills need to be paid to cloud providers depending on the size of the machine.

- ❖ In **server virtualization**, hardware is hidden, and an operating system is run on top of it.
- ❖ **Containerization** is a way to run an app on top of an operating system.
- ❖ **Virtualization** depends on the underlying host operating system but does not worry about the hardware unless it has *enough resources*.

Containerization Layers



- ❖ **Hardware infrastructure:** The foundation of each application is a collection of tangible resources that may be put to productive use. In order for containers to function correctly, these resources must be present. They could be running on a laptop or at one of the many data centers connected to the cloud.
- ❖ **Host operating system:** After the hardware layer comes the next layer, the host operating system. As with the hardware layer, this may be as straightforward as installing Windows or *nix on any personal computer, or it could be handled entirely by a cloud service provider.
- ❖ **Container Engine:** This is where things begin to take an exciting turn: the container engine. Container engines are software installed on top of the host operating system and are responsible for virtualizing the resources required by containerized applications.
- ❖ This layer is the easiest to grasp when Docker is executed on a computer. This layer ensures the container is up and running and manages its overall lifecycle.
- ❖ **Containers/Containerized apps are bits of code that include all of the libraries, binaries, and configuration settings that an app needs to execute.**
- ❖ **Containers** are also known as Docker containers. An application that has been containerized operates as its process in “user space,” which is distinct from the operating system’s kernel.

Cara Kerja Containerization?

- ❖ Dengan teknologi *containerization*, ada lebih sedikit *overhead* selama startup dan tidak perlu menyiapkan quest OS terpisah untuk setiap aplikasi karena **semuanya berbagi kernel OS yang sama**.
- ❖ *Containerization* biasanya digunakan untuk mengemas ***banyak layanan mikro individu*** yang membentuk aplikasi *modern*.
- ❖ Setiap **WADAH** adalah **PAKET PERANGKAT LUNAK** yang dapat dieksekusi, berjalan di atas OS host. Sebuah host dapat mendukung banyak wadah (*puluhan, ratusan, atau bahkan ribuan*) secara bersamaan, seperti dalam kasus arsitektur layanan mikro kompleks yang menggunakan banyak pengontrol pengiriman aplikasi dalam wadah.

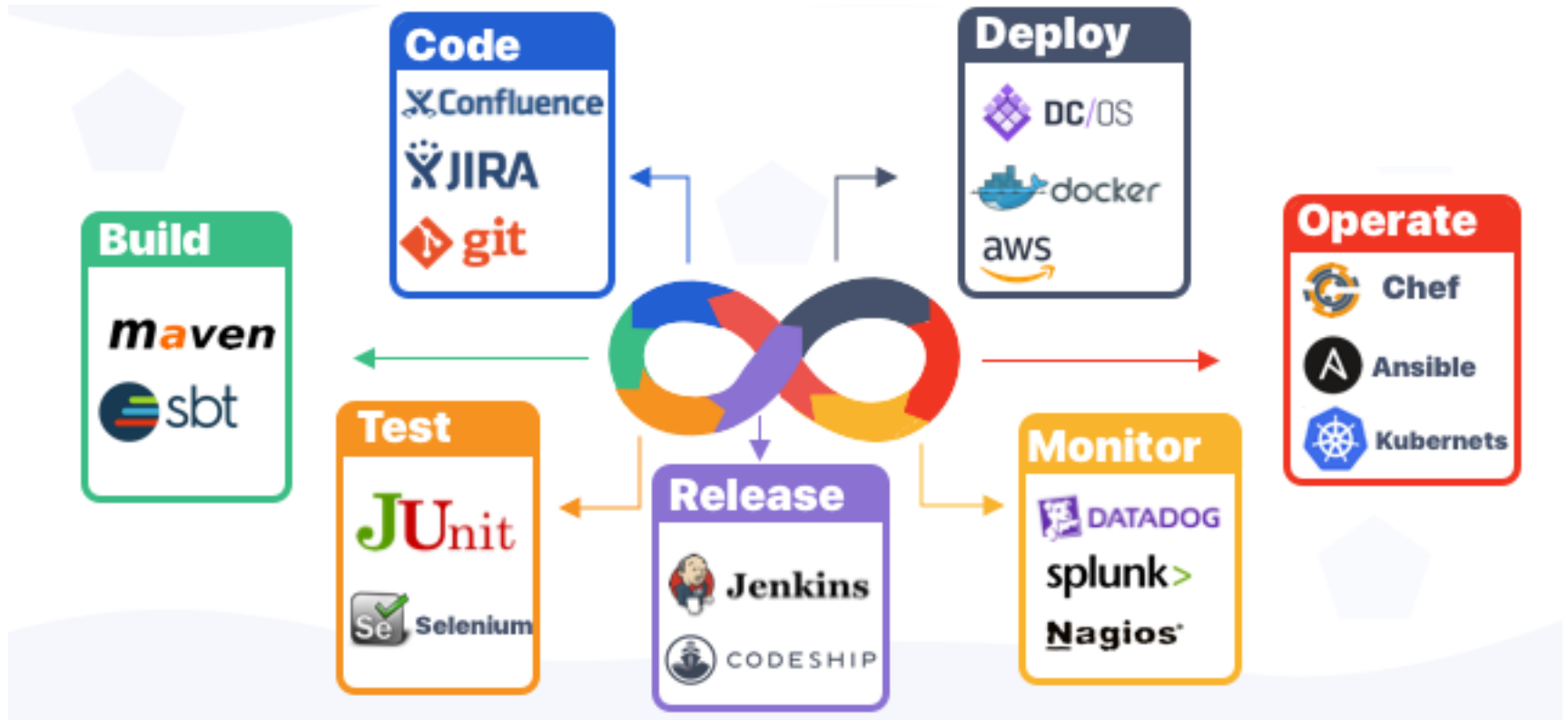
Penyiapan ini berfungsi karena semua container menjalankan proses minimal yang terisolasi sumber daya yang tidak dapat diakses oleh orang lain.

The Benefits of Containerization

Containerization lets the team build the same image locally and in production. **Container apps can reduce situations when something works in one location but not another.** CI/CD pipelines support container construction. These benefits improve team productivity.

Container is portability	What is DevOps dilemma????. People complain about an application functioning in one environment (e.g., staging) but not in another. Usually, the problem is when the environment is difference. Perhaps a dependency was updated. The same container images, including dependencies, can be executed everywhere with containerization.
Container is fast	Containers start faster than virtual machines or bare metal servers. Containers boot in seconds, while virtual machines take minutes, depending on resources and app size.
Container is resource efficient	Containers are more efficient than virtual machines since they only include app-specific files. Virtual machines are gigabytes, while containers are megabytes. Containers let teams use server resources efficiently.
Container is deployment-Development simplicity	✓ Portable containers can be used anywhere. ✓ Containerized apps are fast, small and easy to deploy.

Containerization Tools

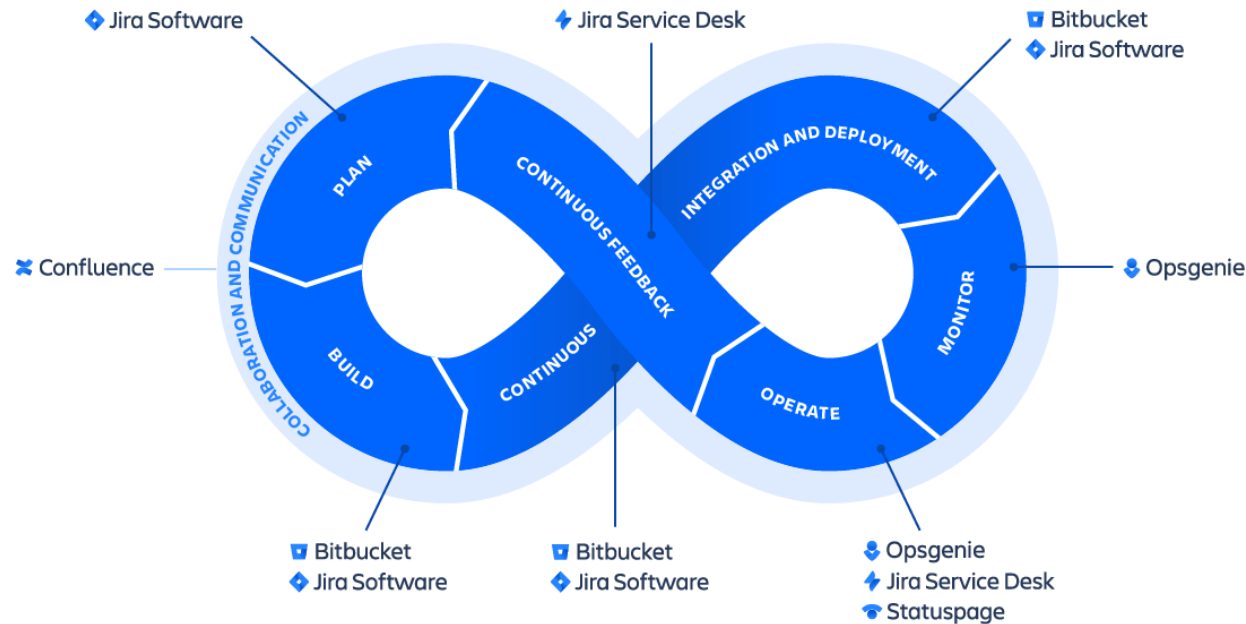


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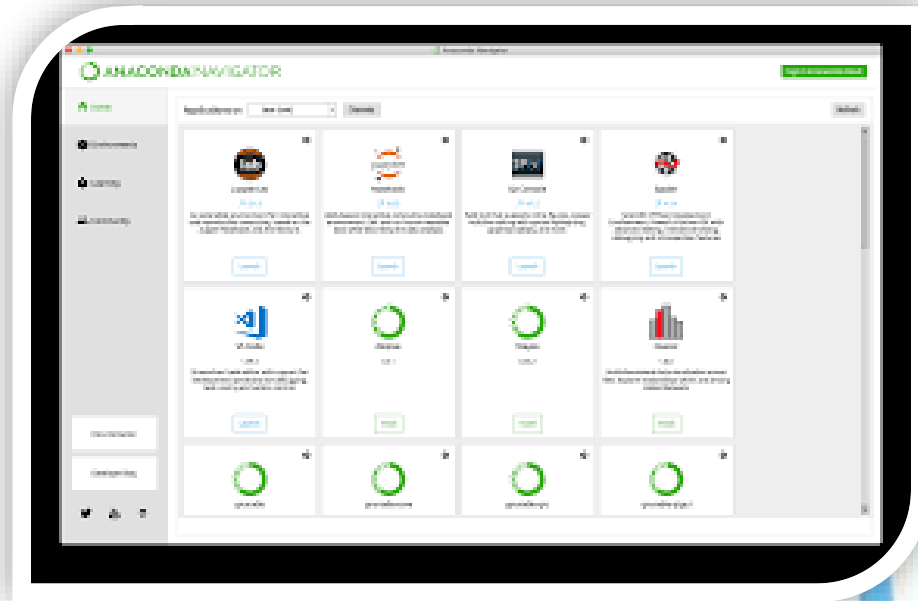
2 - Configuration Management in DevOps

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Configuration Management



Proses ini (*Configuration Management*) berkaitan dengan system engineering yang bertujuan untuk maintain konfigurasi sebuah PROJECT. **Configuration Management** memungkinkan otomatisasi dan standardisasi konfigurasi service/product. Configuration Management lebih mengacu dengan proses semua software yang dipelihara (*maintained*) dan dikonfigurasi (*configured*).

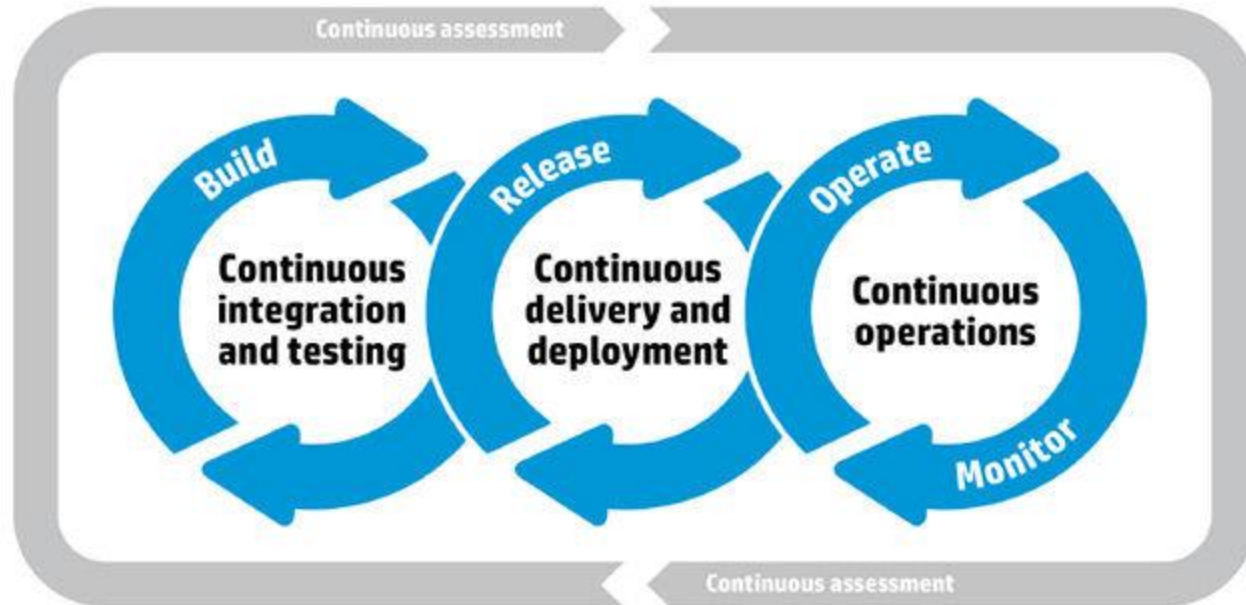


Apakah PYTHON ANACONDA dapat digunakan untuk Configuration Management?



Anaconda adalah sebuah platform yang umumnya digunakan untuk mengelola lingkungan Python dan paket-paketnya dalam pengembangan data ilmiah dan analisis data. Anaconda menyediakan kemudahan dalam membuat dan mengelola lingkungan Python yang terisolasi (*conda environments*) serta menginstal paket-paket Python dan library ilmiah. Anaconda tidak dirancang khusus untuk Configuration Management dalam konteks DevOps.

Configuration Management

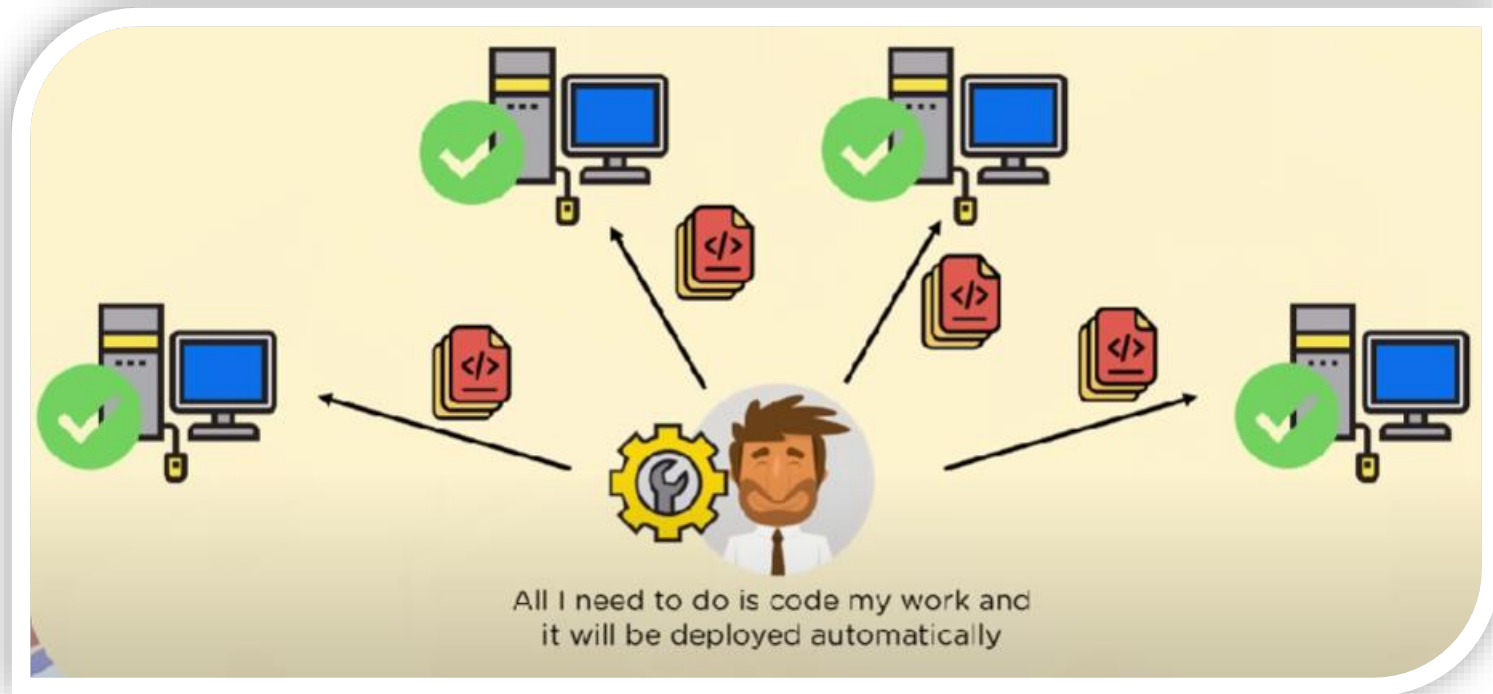


Saya percaya, **masing-masing step di dalam DevOps membutuhkan konfigurasi**. Configuration Management (Manajemen Konfigurasi) adalah salah satu elemen penting dalam praktik DevOps. Ini membantu dalam mengelola konfigurasi target software dan infrastruktur secara konsisten dan efisien selama **development lifecycle** (siklus hidup pengembangan) dan pengiriman perangkat lunak.

Configuration Management?

- ❖ In software development circles, **configuration management is the process by which all environments hosting software are configured and maintained.**
- ❖ **Every development pipeline requires multiple environments** – unit testing, integration testing, acceptance testing, load testing, system testing, end-user testing, etc. These environments become increasingly complex as the testing moves towards pre-prod and prod environments.
- ❖ Configuration management is an automated process ensuring optimal configuration of these environments.
- ❖ The configuration of testing environments is critical for the success of testing teams. The exact configuration makes every resource – servers, networks, data centers, operating systems, IT assets, configuration files – function as they must to facilitate success. These environments must be meticulously managed, and all configuration changes must be tracked to ensure they are traceable.

Configuration management berfungsi untuk menetapkan konfigurasi pada server atau lingkungannya.



Configuration management is the process to
(1) Configuration Identification, (2) Configuration Control and
(3) Configuration Audit. When choosing DevOps Configuration
Management tools, select one that facilitates these three
functions with the most efficiency and ease.

Elements of DevOps Configuration Management

Configuration Identification	We need to identify the configuration of the environment to be maintained. One can also use discovery tools to identify configurations automatically.
Configuration Control	Remember that, it might not remain unchanged once the configuration has been identified. Consequently, there needs to be some mechanism in place to track and control changes to the configuration. Most configuration management frameworks have a change management process regulating these configurational changes.
Configuration Audit	Even with control mechanisms, changes may bypass them. Configuration audits at regular intervals prevent such incidents. When choosing DevOps Configuration Management tools, select one that facilitates these three functions with the most efficiency and ease.

Like DevOps, Configuration Management spans operational and development activities within an organization.

The **PRIMARY COMPONENTS** that comprise comprehensive configuration management are:

- 1) **Artifact repository.** An artifact repository stores machine files – binaries, test data, and libraries. Consider it as a database for infrequently used files within an environment.
- 2) **Source code repositor.** The source code repository contains all versions of the code. In other words, it is the database of source code used by all developers on a team or project.
- 3) **Database for Configuration Management.** Data architecture or a database devoted to DevOps Configuration Management works across different systems and applications related to said management.

Top 5 Configuration Management Tools



ANSIBLE

Ansible



puppet

Puppet



CHEF™

Chef



CFEngine

CFEngine



SALTSTACK

Saltstack

CONFIGURATION MANAGEMENT TOOLS



Terraform

SALTSTACK



puppet



git



CHEF



ANSIBLE



docker



Exercise *for* Students

Cara Install Docker di Windows 10 / 11 Lengkap #tutorialdockere:

<https://www.youtube.com/watch?v=6tU3-s-KCo>

Cara menjalankan laravel dengan docker untuk pemula:

<https://www.youtube.com/watch?v=BcF9FW8EzgU>

Containerization *in* DevOps

- 1) Apa yang dimaksud dengan containerisasi, dan bagaimana containerisasi mendukung prinsip-prinsip DevOps dalam pengembangan dan pengiriman software?
- 2) Bagaimana containerisasi membantu dalam mengisolasi aplikasi dan dependensinya, dan mengapa ini penting dalam pengembangan software/sistem di era aplikasi modern?
- 3) Bagaimana proses ***develop*** dan ***deploy*** container berbeda dari pendekatan tradisional dalam pengembangan software?
- 4) Apa keuntungan dari penggunaan container untuk mengelola aplikasi di berbagai lingkungan, seperti lingkungan *development*, *testing* dan *production*?
- 5) Bagaimana containerisasi membantu dalam mengatasi masalah ketergantungan dan konflik **KONFIGURASI** saat mengimplementasikan perubahan dalam aplikasi/sistem?

Configuration Management *in* DevOps

- 1) Apa yang dimaksud dengan Configuration Management dalam konteks DevOps, dan mengapa hal ini penting untuk kelancaran pengembangan dan pengiriman sebuah project sistem/software?
- 2) Bagaimana Configuration Management membantu dalam menjaga konsistensi dan keterukuran dalam lingkungan *development*, *testing*, dan *production*?
- 3) Bagaimana tim DevOps dapat mengelola infrastruktur sebagai kode dengan menggunakan Configuration Management, dan apa manfaatnya?
- 4) Bagaimana Configuration Management membantu dalam otomatisasi tugas-tugas pengelolaan konfigurasi, dan mengapa otomatisasi ini penting dalam praktik DevOps?
- 5) Apa peran Configuration Management dalam mendukung pengiriman software berkelanjutan (*continuous delivery*) dan manajemen perubahan dalam siklus hidup sistem/software?



**Buatlah chatbot sederhana
menggunakan Lovable AI dan
Gemini 2.0 Flash?**

#1

Buat Google Gemini API Key

Gemini API Pricing

Gemini API pricing | Google AI for Developers

Link: <https://ai.google.dev/pricing>

Saya hanya menggunakan versi free dari Gemini 1.5 Flash API

Google AI for Developers

Models

Solutions

Code assistance

Showcase

Community

Search

Gemini API docs

API Reference

SDKs

Pricing

Cookbook

Pricing models

Priced to help you bring your app to the world

Gemini 1.5 Flash

Gemini 1.5 Flash-8B

Gemini 1.5 Pro

Gemini 1.0 Pro

Text Embedding 004

Gemini 1.5 Flash

AVAILABLE NOW

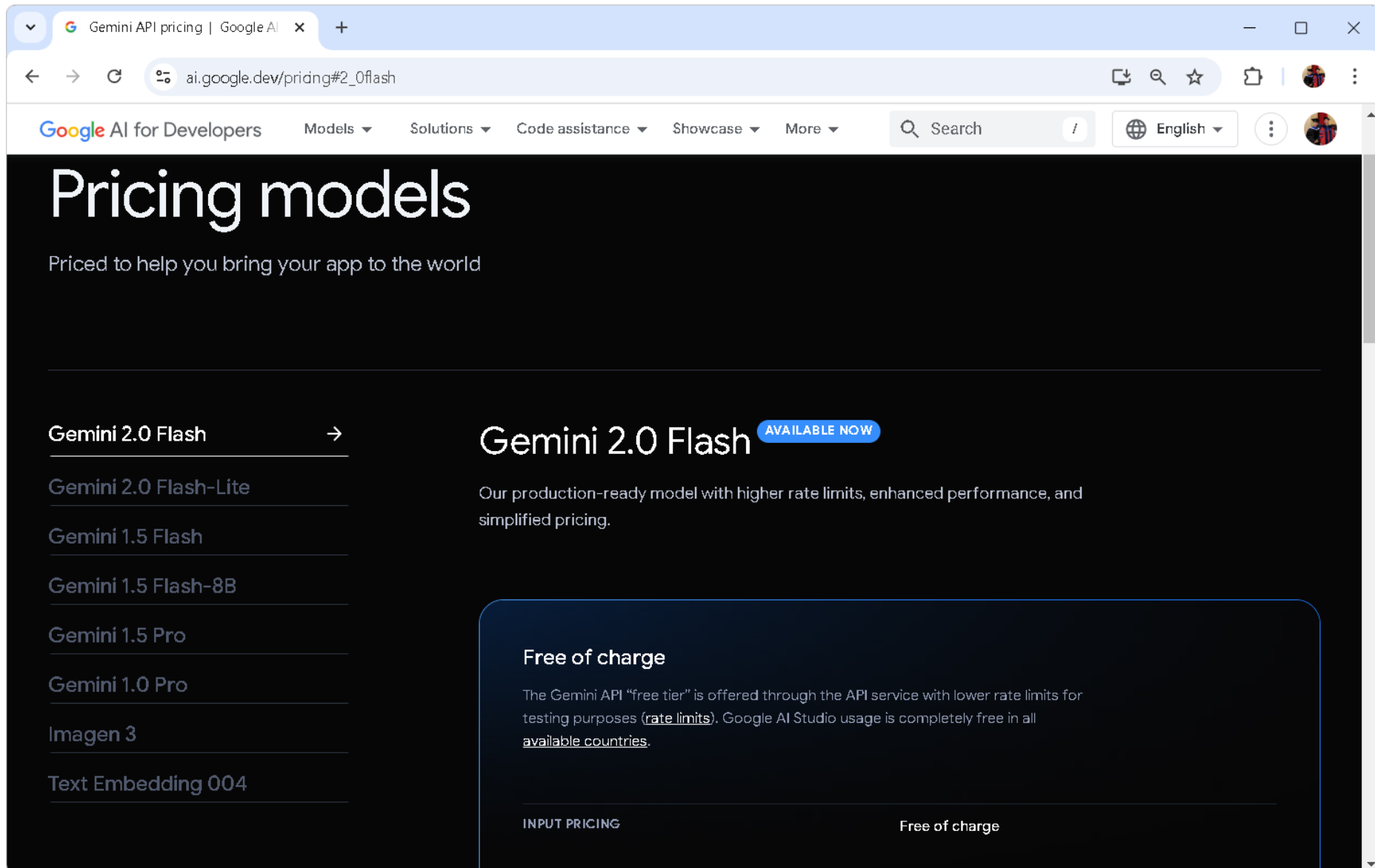
Our fastest multimodal model with great performance for diverse, repetitive tasks and a 1 million context window. Now generally available for production use.

Free of charge

The Gemini API "free tier" is offered through the API service with lower rate limits for testing purposes. Google AI Studio usage is completely free in all [available countries](#).

RATE LIMITS	
	15 RPM (requests per minute)
	1 million TPM (tokens per minute)
	1,500 RPD (requests per day)

Google Gemini 2.0 Flash (New Model)



The screenshot shows a web browser window with the URL `ai.google.dev/pricing#2_0flash`. The page is titled "Pricing models" and features a sidebar with links to various models: Gemini 2.0 Flash, Gemini 2.0 Flash-Lite, Gemini 1.5 Flash, Gemini 1.5 Flash-8B, Gemini 1.5 Pro, Gemini 1.0 Pro, Imagen 3, and Text Embedding 004. The main content area highlights the Gemini 2.0 Flash model with an "AVAILABLE NOW" badge. It describes the model as production-ready with higher rate limits and simplified pricing. A "Free of charge" section explains that the API's "free tier" is available through the API service with lower rate limits for testing purposes, while Google AI Studio usage is completely free in all available countries. At the bottom, a table shows the input pricing for the Gemini 2.0 Flash model as "Free of charge".

Gemini API pricing | Google AI

ai.google.dev/pricing#2_0flash

Google AI for Developers Models Solutions Code assistance Showcase More Search English

Pricing models

Priced to help you bring your app to the world

- Gemini 2.0 Flash →
- Gemini 2.0 Flash-Lite
- Gemini 1.5 Flash
- Gemini 1.5 Flash-8B
- Gemini 1.5 Pro
- Gemini 1.0 Pro
- Imagen 3
- Text Embedding 004

Gemini 2.0 Flash AVAILABLE NOW

Our production-ready model with higher rate limits, enhanced performance, and simplified pricing.

Free of charge

The Gemini API “free tier” is offered through the API service with lower rate limits for testing purposes ([rate limits](#)). Google AI Studio usage is completely free in all [available countries](#).

INPUT PRICING	Free of charge
---------------	----------------

Get Gemini API key

Go to: <https://ai.google.dev/gemini-api/docs/api-key>

Click di button “**Get a Gemini API key in Google AI Studio**”.

The screenshot shows the Google AI for Developers website. The browser address bar displays the URL `ai.google.dev/gemini-api/docs/api-key`. The navigation bar includes links for Google AI for Developers, Models, Solutions, Code assistance, Showcase, and Community. A blue sidebar on the left contains links for Gemini API docs, API Reference, SDKs, Pricing, and Cookbook. The main content area features a blue banner for Gemini 2.0 Flash Experimental, a breadcrumb trail (Home > Gemini API > Models), and the title 'Get a Gemini API key'. Below the title, a paragraph explains that an API key is needed to use the Gemini API and can be created in Google AI Studio. A blue button labeled 'Get a Gemini API key in Google AI Studio' is highlighted with a red checkmark. The section 'Set up your API key' follows, with text explaining that for initial testing, an API key can be hard-coded, but it should be temporary and replaced by a local environment variable for security.

Overview

Get started

- Quickstart
- API keys
- Libraries
- Release notes
- Developer forum

Models

- Gemini
- Gemini 2.0
- Experimental models

Capabilities

- Text generation
- Vision

Gemini 2.0 Flash Experimental is now available! [Learn more](#)

Home > Gemini API > Models

Get a Gemini API key

To use the Gemini API, you need an API key. You can create a key with a few clicks in Google AI Studio.

[Get a Gemini API key in Google AI Studio](#)

Set up your API key

For initial testing, you can hard code an API key, but this should only be temporary since it is not secure. The rest of the page discusses how to set up your API key locally as an environment variable with different operating systems.

Create Gemini API Key

Google AI Studio: <https://aistudio.google.com/app/apikey>

Click button “**Create API Key**” untuk membuat API key yang baru. Disini saya sudah membuat sebuah API key yang akan digunakan pada CURL.

Google AI Studio Get API key

Get API key

API keys

Quickly test the Gemini API

API quickstart guide

```
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key=GEMINI_API_KEY" \
-H 'Content-Type: application/json' \
-X POST \
-d '{
  "contents": [{
    "parts": [{"text": "Explain how AI works"}]
  }]
}'
```

Use code with caution.

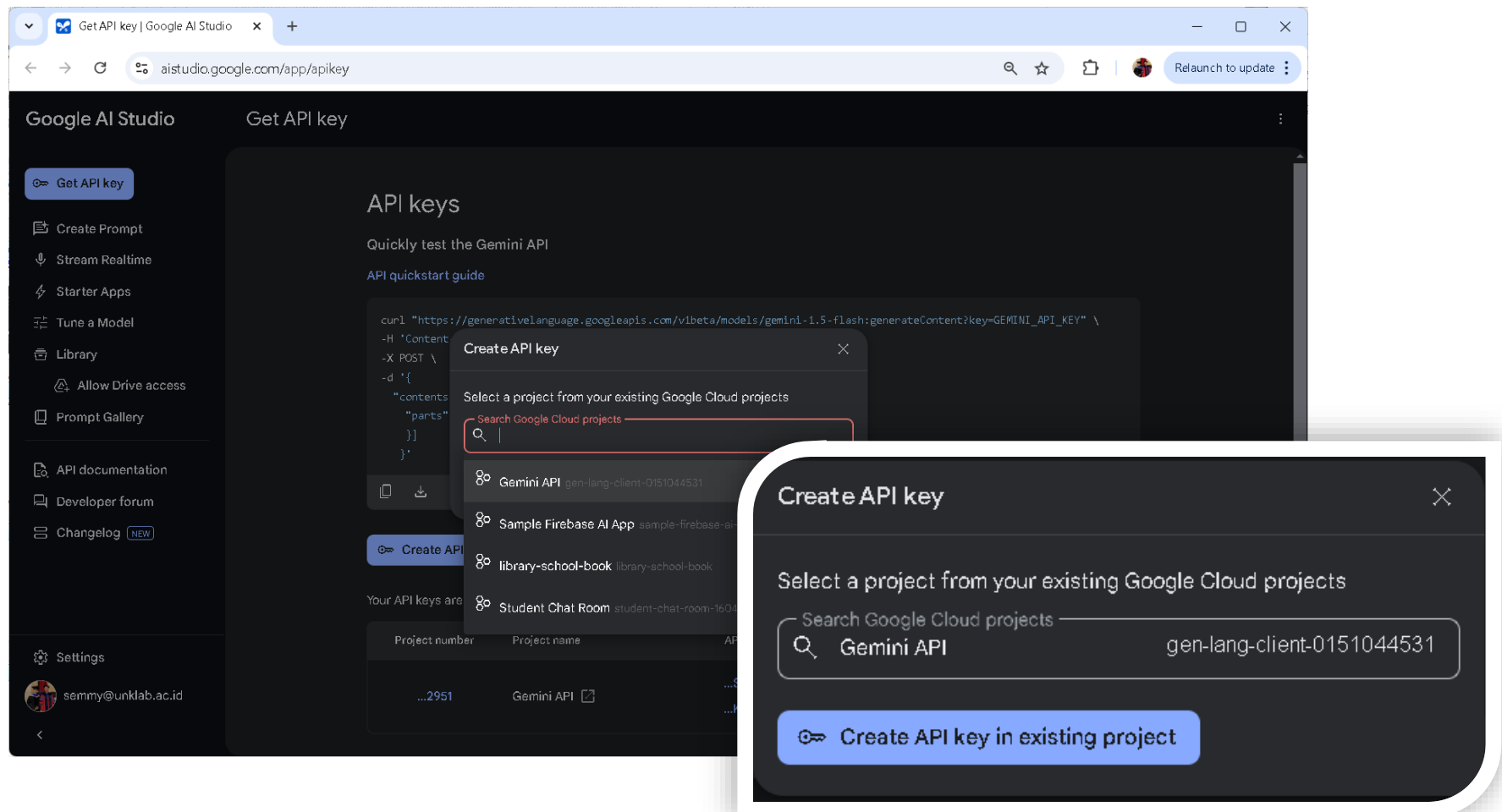
Create API key

Your API keys are listed below. You can also view and manage your project and API keys in Google Cloud.

Project number	Project name	API key	Created	Plan
...2951	Gemini API	...K9YY	Dec 29, 2024	Free of charge Set up Billing View usage data

Search Google Cloud projects

Select a project from your existing Google Cloud projects. Pilih Opsi “**Gemini API**”. Selanjutnya, click button “**Create API key in existing project**”.



The screenshot shows the Google AI Studio 'Get API key' page. The left sidebar contains navigation links: Get API key, Create Prompt, Stream Realtime, Starter Apps, Tune a Model, Library, Allow Drive access, Prompt Gallery, API documentation, Developer forum, and Changelog. The main content area is titled 'API keys' and includes a 'Quickly test the Gemini API' section with a cURL command and an 'API quickstart guide' link. A 'Create API key' modal is open, displaying a search bar for 'Search Google Cloud projects' and a list of projects. The 'Gemini API' project is selected, and the 'Create API key in existing project' button is highlighted.

API keys

Quickly test the Gemini API

API quickstart guide

```
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key=GEMINI_API_KEY" \
-H "Content-Type: application/json" \
-X POST \
-d '{
  "contents": [
    {
      "parts": [
        {
          "text": "Hello, Gemini!"
        }
      ]
    }
  ]
}'
```

Create API key

Select a project from your existing Google Cloud projects

Search Google Cloud projects

Gemini API gen-lang-client-0151044531

Sample Firebase AI App sample-firebase-ai-app

library-school-book library-school-book

Student Chat Room student-chat-room-1604

Create API key

Select a project from your existing Google Cloud projects

Search Google Cloud projects

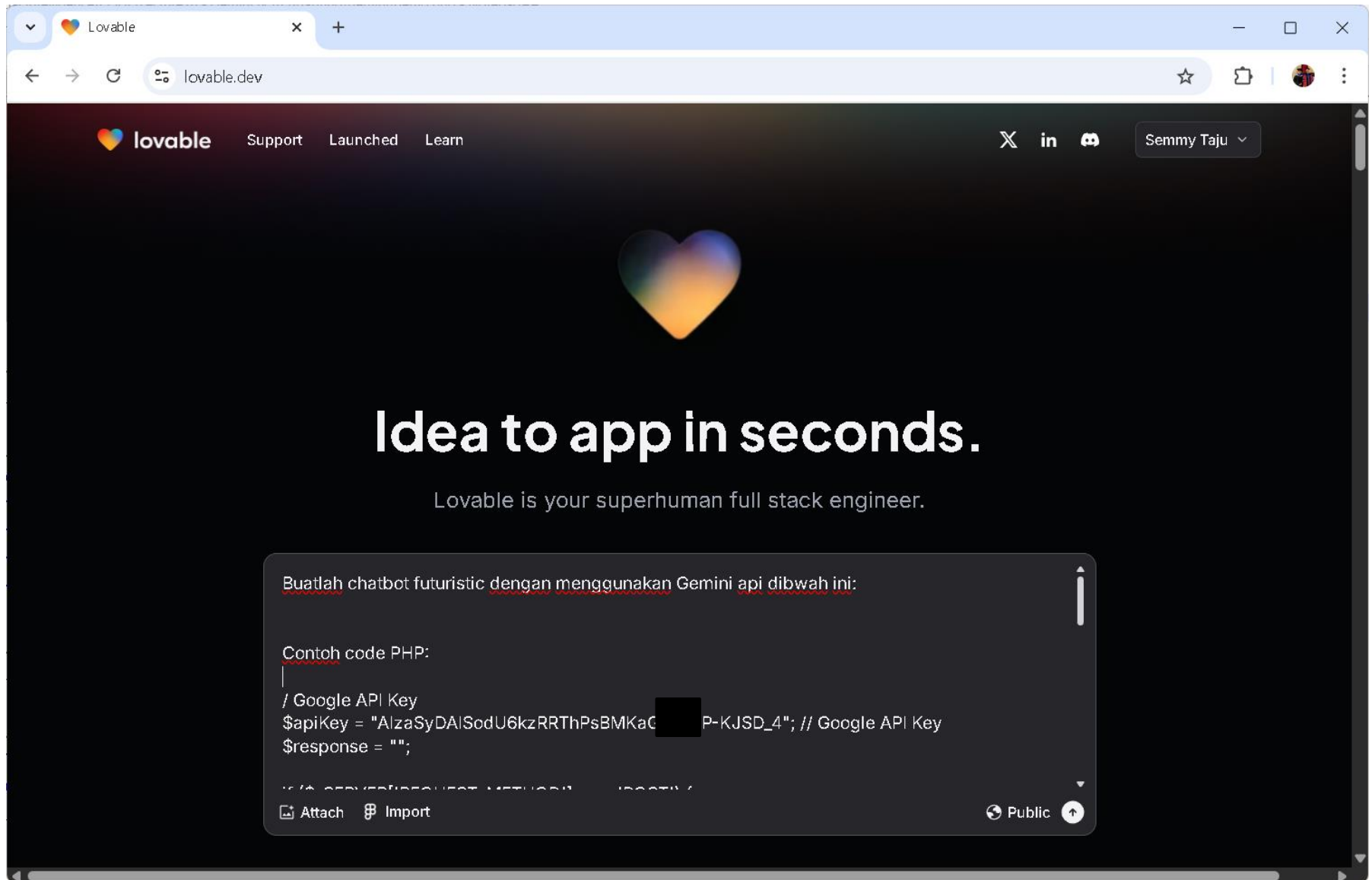
Gemini API gen-lang-client-0151044531

Create API key in existing project

#2

**Connect Lovable AI & Gemini
API Key using Prompt**

1. Lovable input Prompt



Tambahkan di Prompt:

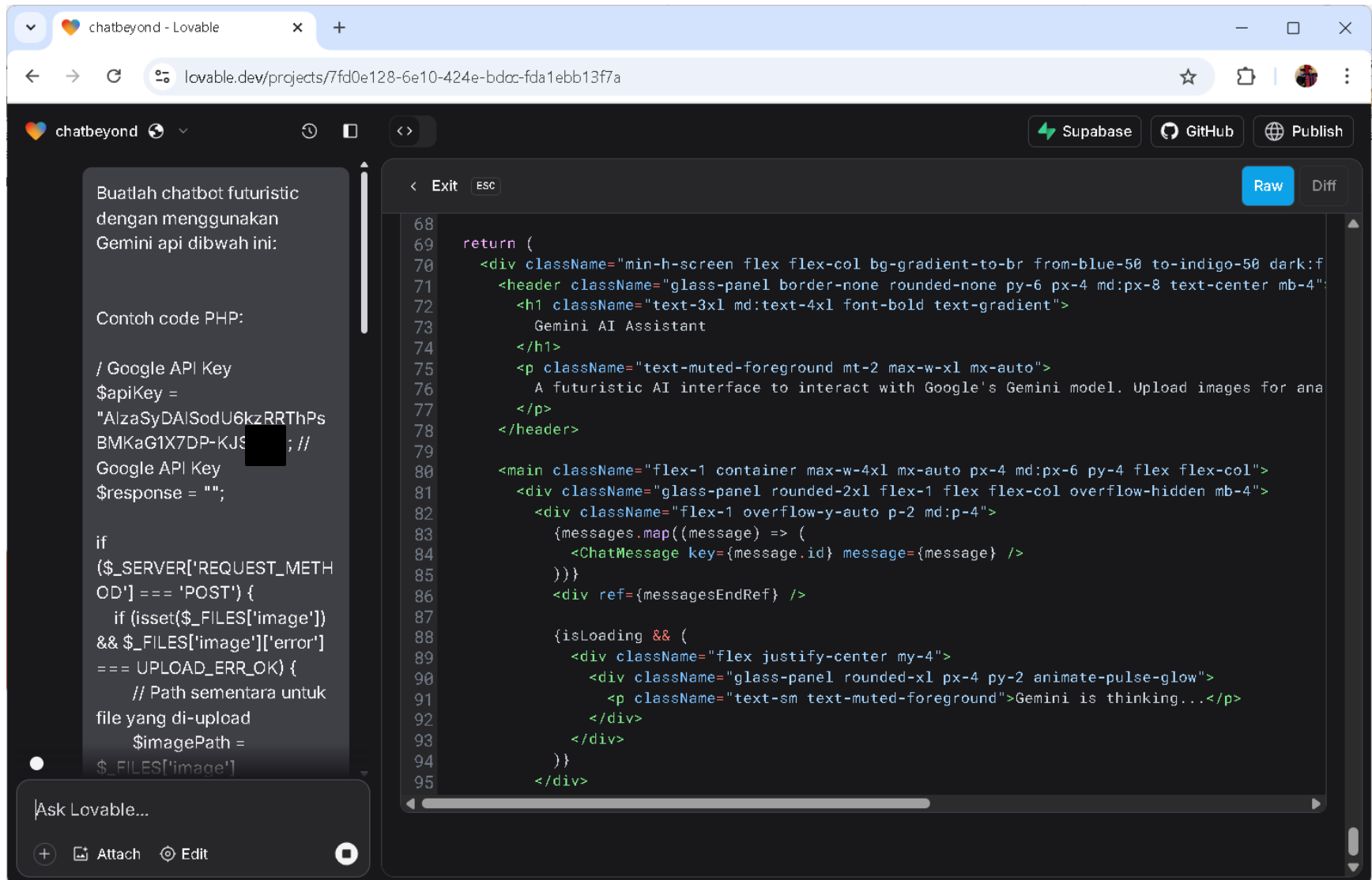
// Google API Key

```
$apiKey = "AlzaSyDAsISodUe6kzRRThPsBMKaGs1X7DP-KJSD_4"; // Google API Key
```

// URL endpoint API

```
$url =  
"https://generativelanguage.googleapis.com/v1beta/models/ge  
mini-1.5-flash:generateContent?key={$apiKey}";
```

2. Lovable AI will generate your code.



The screenshot shows the Lovable AI web interface. On the left, a chat window contains a prompt in Indonesian: "Buatlah chatbot futuristic dengan menggunakan Gemini api dibawah ini:" followed by "Contoh code PHP:". The code is a PHP script that uses the Google Gemini API to interact with a chatbot. It includes a header section with a title "Gemini AI Assistant" and a main content area that displays chat messages and a loading indicator. The right side of the interface shows the generated code in a dark-themed editor. The code is a PHP script that uses the Google Gemini API to interact with a chatbot. It includes a header section with a title "Gemini AI Assistant" and a main content area that displays chat messages and a loading indicator. The code is syntax-highlighted and includes line numbers. At the bottom of the chat window, there is an input field labeled "Ask Lovable..." and buttons for "Attach" and "Edit".

chatbeyond - Lovable

lovable.dev/projects/7fd0e128-6e10-424e-bd0c-fda1ebb13f7a

chatbeyond

Buatlah chatbot futuristic dengan menggunakan Gemini api dibawah ini:

Contoh code PHP:

```
/ Google API Key
$apiKey =
"AlzaSyDAISodU6kzRRThPs
BMKaG1X7DP-KJS "; //
Google API Key
$response = "";

if
($_SERVER['REQUEST_METH
OD'] === 'POST') {
    if (isset($_FILES['image'])
    && $_FILES['image']['error']
    === UPLOAD_ERR_OK) {
        // Path sementara untuk
        file yang di-upload
        $imagePath =
        $_FILES['image']
```

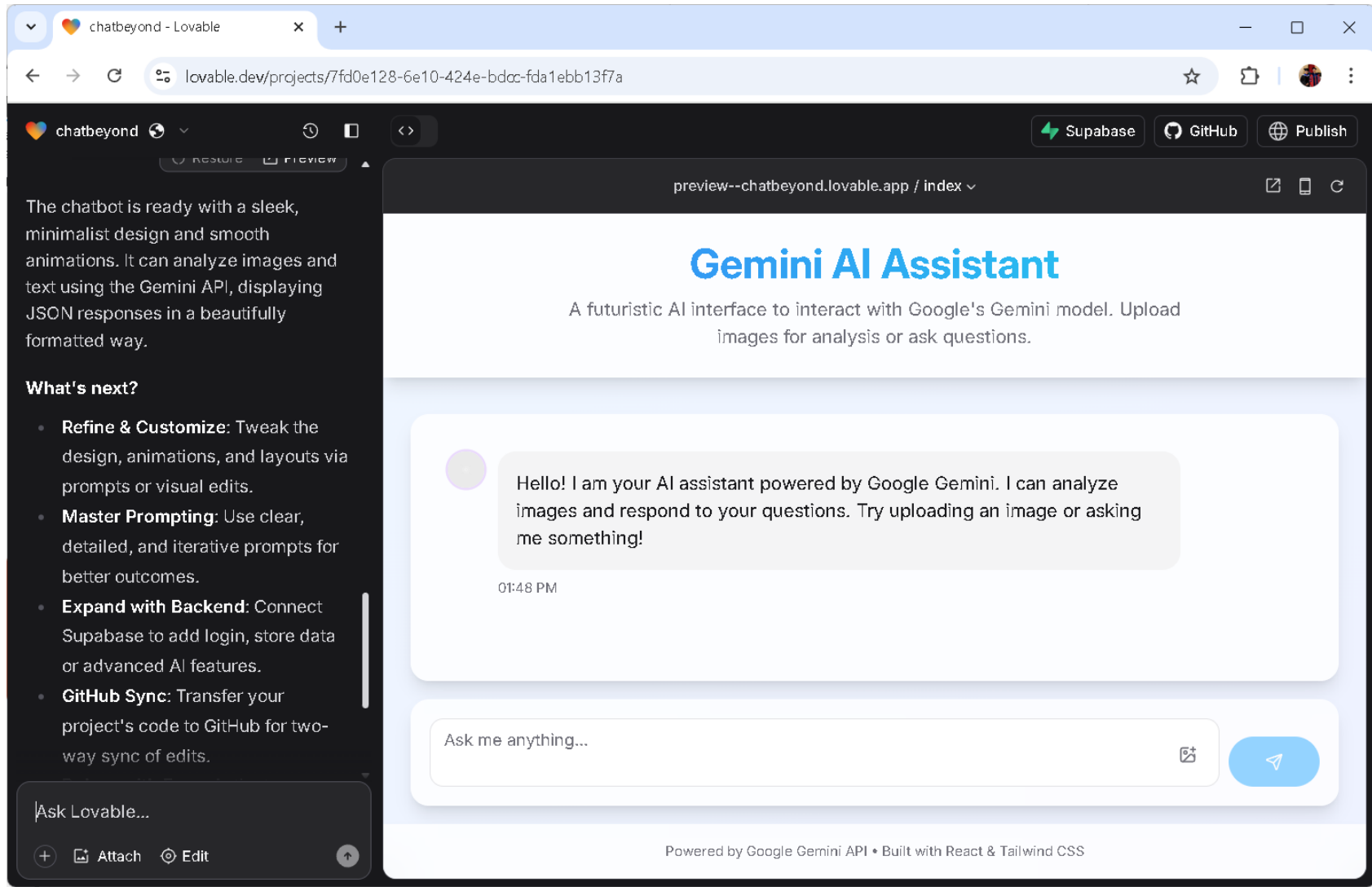
Raw Diff

```
68
69 return (
70     <div className="min-h-screen flex flex-col bg-gradient-to-br from-blue-50 to-indigo-50 dark:f
71     <header className="glass-panel border-none rounded-none py-6 px-4 md:px-8 text-center mb-4"
72     <h1 className="text-3xl md:text-4xl font-bold text-gradient">
73         Gemini AI Assistant
74     </h1>
75     <p className="text-muted-foreground mt-2 max-w-xl mx-auto">
76         A futuristic AI interface to interact with Google's Gemini model. Upload images for ana
77     </p>
78 </header>
79
80 <main className="flex-1 container max-w-4xl mx-auto px-4 md:px-6 py-4 flex flex-col">
81     <div className="glass-panel rounded-2xl flex-1 flex flex-col overflow-hidden mb-4">
82         <div className="flex-1 overflow-y-auto p-2 md:p-4">
83             {messages.map((message) => (
84                 <ChatMessage key={message.id} message={message} />
85             ))}
86             <div ref={messagesEndRef} />
87
88             {isLoading && (
89                 <div className="flex justify-center my-4">
90                     <div className="glass-panel rounded-xl px-4 py-2 animate-pulse-glow">
91                         <p className="text-sm text-muted-foreground">Gemini is thinking...</p>
92                     </div>
93                 </div>
94             )}
95         </div>
```

Ask Lovable...

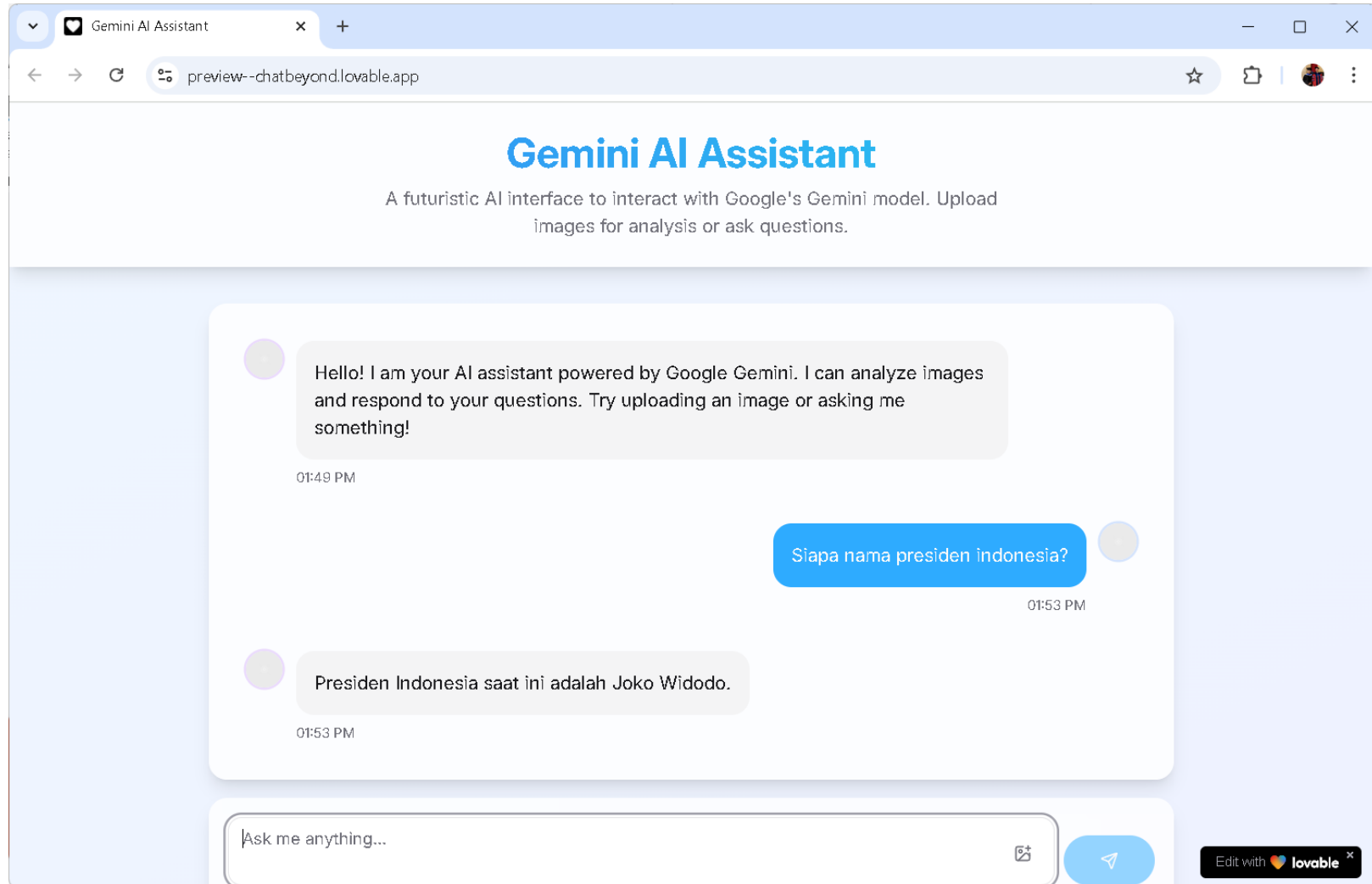
+ Attach Edit

3. Your chatbot is ready !!!



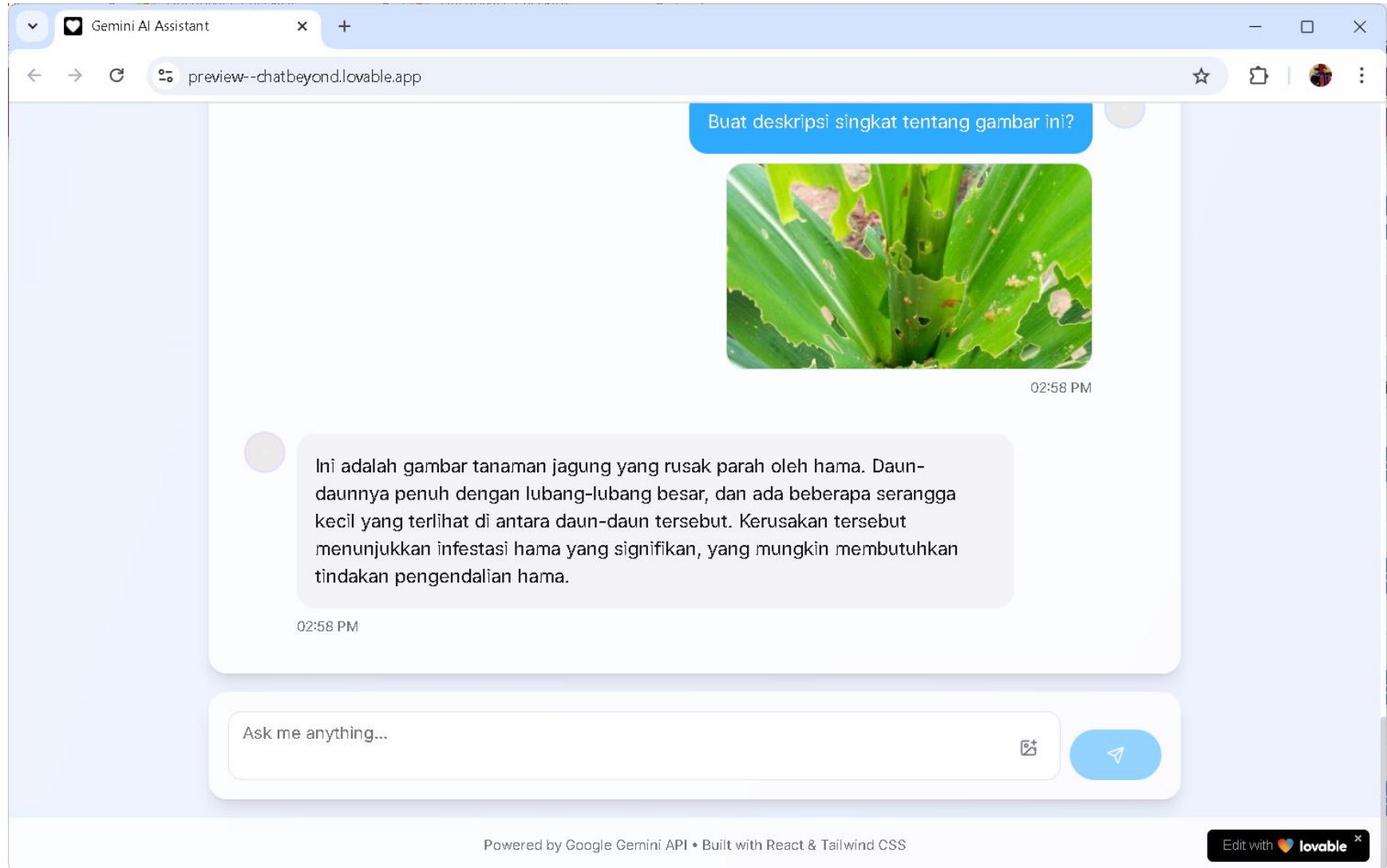
3. Chatbot can be accessed publicly.

<https://preview--chatbeyond.lovable.app/>



3. Chatbot can be accessed publicly.

<https://preview--chatbeyond.lovable.app/>



END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

